



Anas TBER

Data Science & AI Engineering Student — Centrale Lyon

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Engineering student at Centrale Lyon with hands-on experience in graph-based ML, NLP, and large-scale data engineering. Seeking an **apprenticeship (alternance) in Data Science or AI** starting **September 2026**.

Education

- École Centrale de Lyon** — *General Engineering* Lyon · Sep. 2023 – Apr. 2027
- **Courses:** Machine Learning & Applications, Collaborative Algorithms, Probability & Stochastic Processes, Algorithms for Reasoning, OOP, Enterprise Management
- Lycée Henri Loritz** — *CPGE, PCSI / PSI** Nancy · Sep. 2020 – Jun. 2023
- Intensive preparatory classes in Physics, Mathematics & Engineering Sciences for France's top engineering school entrance exams (Grandes Écoles).

Experience

- Data Scientist Intern** — **BNP Paribas CIB** — *Compliance, Anti-Money Laundering* Paris · Jan. 2026 – Jul. 2026
- Delivered a **regulator-approved** AML fraud-detection system covering the full cycle from raw data ingestion to compliance dashboards, operating on a **30M+ entity** transaction network
 - Modeled transaction networks using graph algorithms (BFS, DFS, Union-Find) for anomaly detection; designed a **cycle detection** system to identify money mule networks and circular transaction flows
 - Engineered graph-based features (degree centrality, cycle membership, neighbor risk scores) to surface high-risk entities across multiple watchlists (Sayari, WikiRan, Crypto, bank sensitivity)
 - Processed and validated **100M+ financial records** ensuring data integrity and regulatory compliance using **Polars** and **PyArrow**; built an interactive **Streamlit** dashboard for compliance analysts
- Tech & Operations Intern** — **Gentle Mates** — *Professional Esports (Squeezie, Gotaga)* Paris · Apr. 2025 – Nov. 2025
- Built an automated social media analytics dashboard (Streamlit/Python) integrating computer vision (**YOLO**) for sponsor logo detection and NLP (**Regex**) for brand mention tracking, improving partnership ROI visibility

Academic & Personal Projects

- Survival Analysis** — **Acute Myeloid Leukemia** — *Challenge Data ENS by QRT* 2025
- **LightGBM** survival model achieving **0.7745 IPCW-C-index** for patient risk stratification on multi-center clinical data; applied feature engineering on genomic datasets with optimized hyperparameters for censored data
- Transformer Architectures & NLP** — *Research Project* 2024
- Sentiment analysis with **CamemBERT** — **95.91% accuracy** on French reviews; studied attention mechanisms, Word2Vec/GloVe/BERT embeddings, and Transformer vs. **Mamba** (SSM) complexity trade-offs
- Ant Colony Optimization** — **Shortest Path Solver** 2024
- Implemented **ACO** metaheuristic in Python (OOP) with pheromone trails and probabilistic decision-making; built a real-time **Tkinter** visualization of pathfinding convergence

Skills

Programming	Python, JavaScript,
AI & Data Science	Pandas, NumPy, Scikit-Learn, PyTorch, LightGBM, LangChain, Jupyter
Data Engineering	Polars, PyArrow, NetworkX, SQL, PostgreSQL
Web & Viz	Streamlit, React, HTML/CSS
DevOps	Git, GitHub, GitLab
Languages	French (Native), Arabic (Native), English (Fluent), Spanish (Beginner)